

Water Garbage Collector

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Introduction

This project presents a floating water-cleaning robot designed to remove floating waste from ponds, lakes, and rivers. The system uses DC motors for movement and a conveyor mechanism for collecting debris from the water surface.

Key Highlights

- Low-cost robotic system for cleaning floating waste
- Conveyor mechanism for debris collection
- Initial concept included ultrasonic and IR sensors
- Current version operates through manual Bluetooth control (HC-05)
- Built on a waterproof floating platform for stability
- Powered through a regulated rechargeable battery

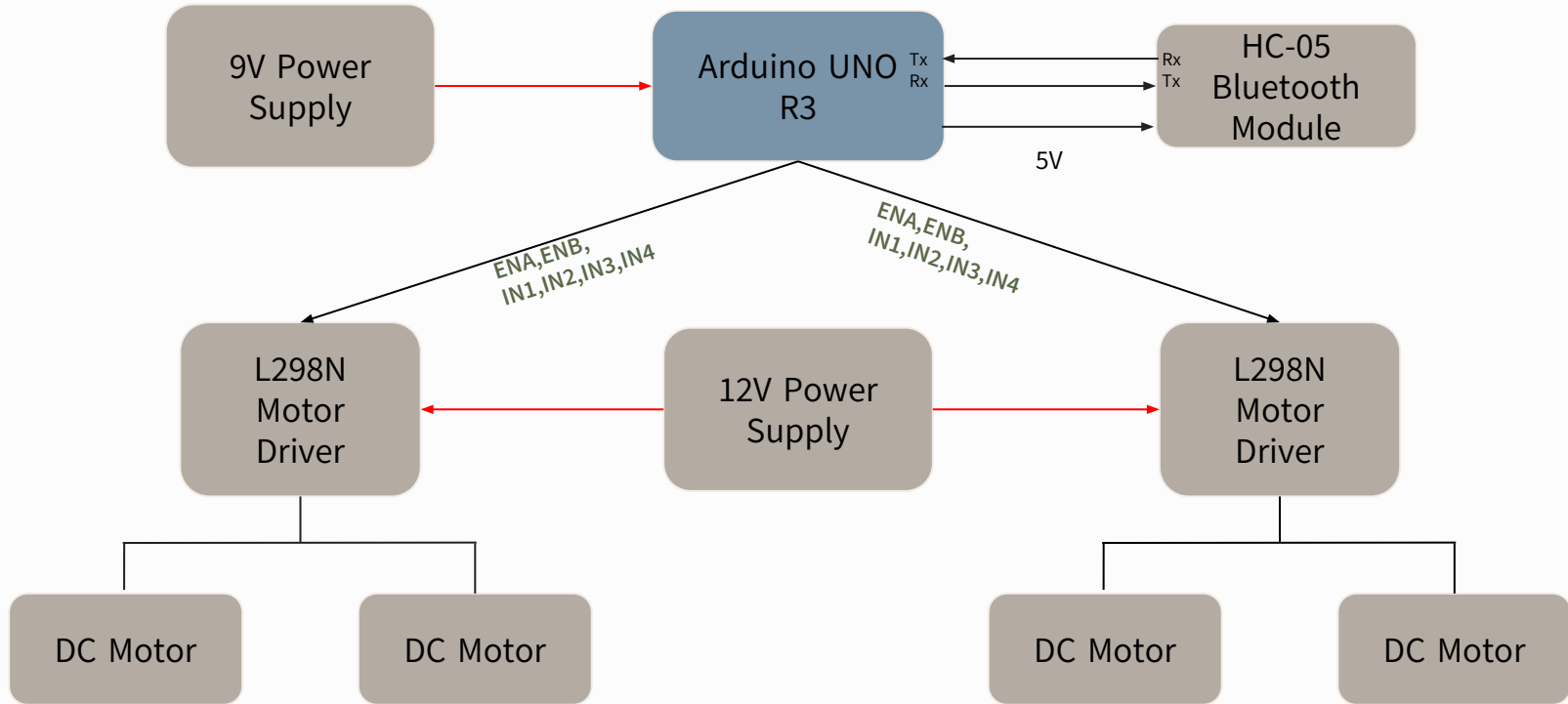
Literature review

- Studies show that floating waste like plastics and thermocol remains on water surfaces for long periods, causing ecological and visual pollution.
- Research highlights that manual cleaning methods are common but inefficient, labour-intensive, and unsafe for workers.
- Existing automated or robotic cleaning systems are effective but often too costly and complex for small or rural water bodies.
- Engineering literature notes that multi-hull floating structures provide better stability for water-based mechanisms.
- Mechanical studies show that conveyor-style collection systems can lift lightweight floating debris effectively when used at suitable angles.
- Overall, the literature points to a gap for simple, low-cost, and stable waste-collection mechanisms for small water bodies.

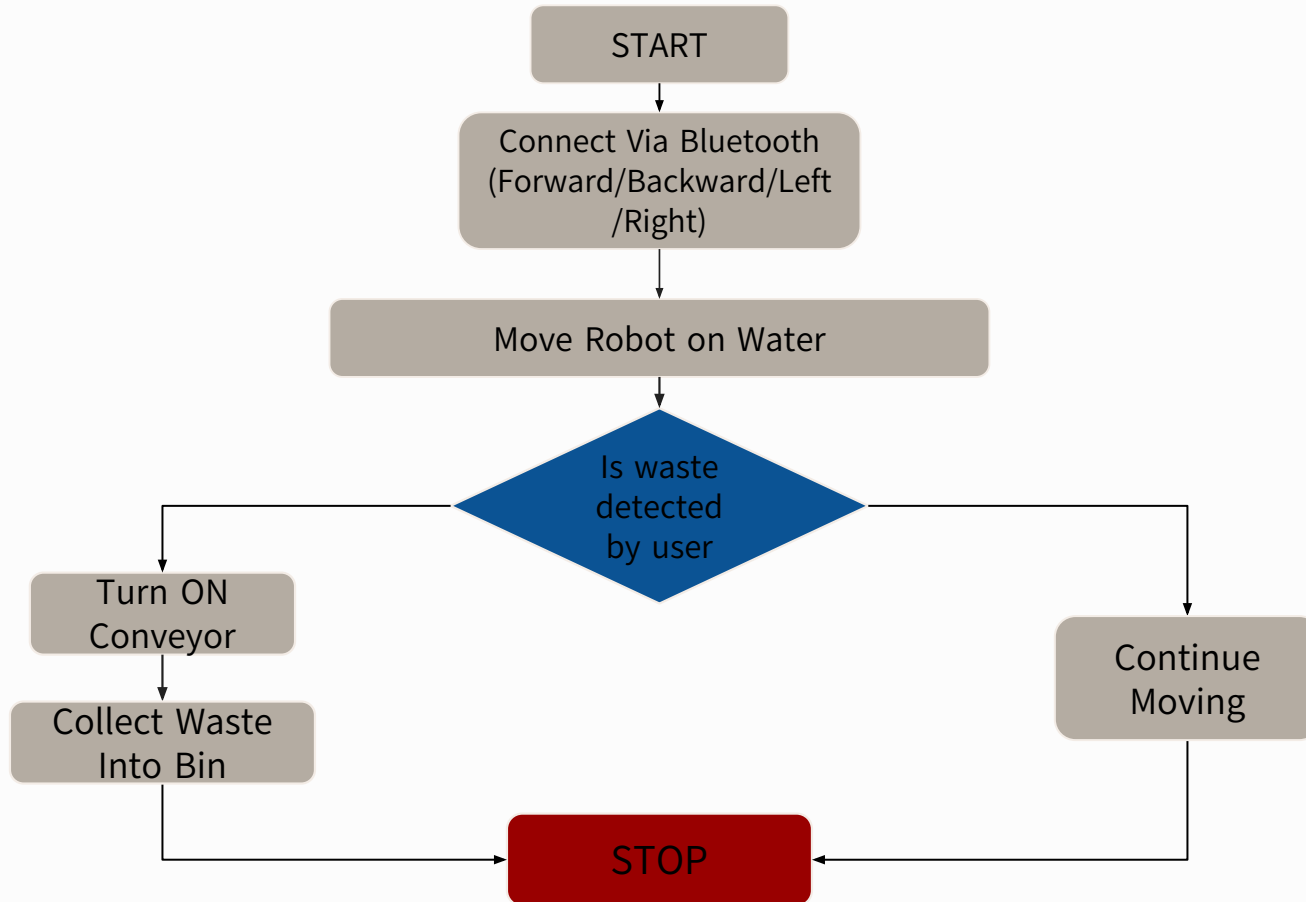
Objective

- Design a floating robot capable of cleaning surface waste efficiently
- Implement a conveyor system to collect and store debris
- Ensure stable, waterproof operation in varying water conditions
- Reduce human effort through semi-automated control
- Deliver an eco-friendly, cost-effective cleaning solution

Block Diagram of Water Garbage Collector



Flow Chart of Water Garbage Collector



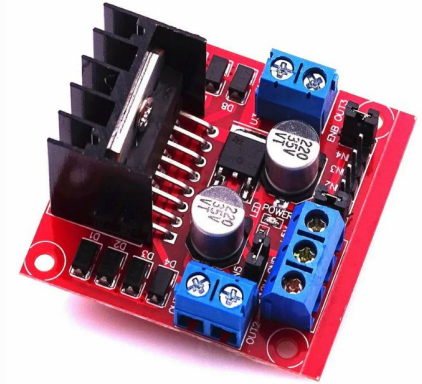


Components & Specifications



L298N Motor Driver Module

Specification		Details
Driver Model		L298N 2A
Driver Chip		Double H Bridge L298N
Motor Supply Voltage (Maximum)		46V
Motor Supply Current (Maximum)		2A
Logic Voltage		5V
Driver Voltage		5-35V
Driver Current		2A
Logical Current		0-36mA
Maximum Power (W)		25W
Current Sense for Each Motor		Yes
Heatsink		Yes
Power-On LED Indicator		Yes
Weight (gm)		25
Dimensions (mm)		44 x 44 x 28 (L x B x H)



DC Motor

RPM	100
Voltage	3-12V
Output Torque	3.5 kg-cm
Weight	200gm



HC-05 Bluetooth Module

Serial communications	9600-115200bps
Frequency	2.4~2.524 GHz
Power Supply	5VDC
Operating Current	30mA
Range	<100m
Dimensions	30mm x 14mm x 2.2 mm
Weight	5 grams



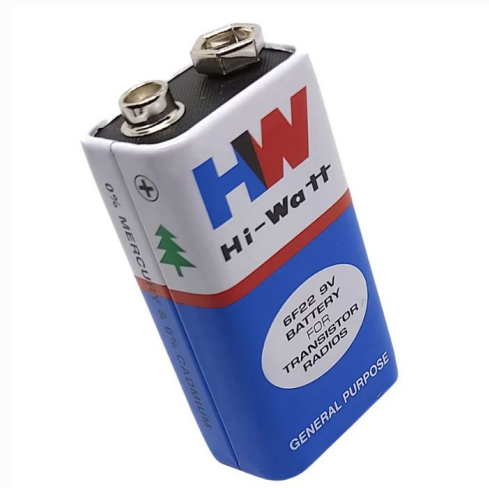
Battery Holder (Two for 12V)

Compatible Battery Size	AA Battery
Material	Plastic
Internal Connections	Series
Cable Length (cm)	15cms
Dimensions	62 x 58 x 15mm
Weight	15 grams



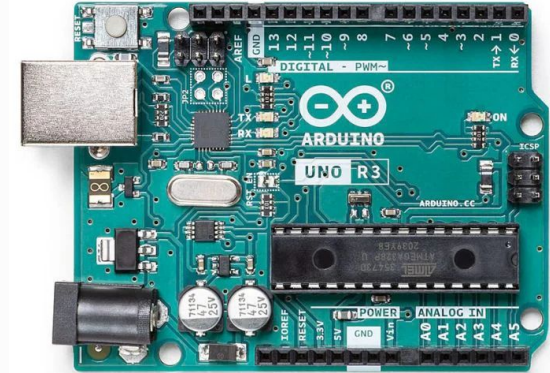
9V Battery

Discharge Time	270hm, 9hours
Nominal Voltage (V)	9
Jacket Material	Metal
Discharge Terminal Type	Press Stud Terminal
Dimensions (mm) LxWxH	4.5 x 2.2 x 1.5
Weight (gm)	178



Arduino UNO R3

Arduino Model	Uno R3
Microcontroller	ATmega328
Operating Voltage	5V
Supply Voltage (recommended)	7-12V
Maximum supply voltage	20V
Digital I/O Pins	14 (of which 6 provide PWM output)
Analog Input Pins	6
DC Current per I/O Pin	40mA
DC Current for 3.3V Pin	50mA
Flash Memory	32 KB of which 0.5 KB is used by the bootloader
SRAM	2KB
EEPROM	1KB
Clock Speed	16MHz



Work Done

- Designed a floating waterproof body to support all components on water.
- Installed DC motors for propulsion and connected them using the **L298N motor driver**.
- Built a working **conveyor mechanism** to lift and collect floating waste.
- Controlled the conveyor motor using the **L298N motor driver**.
- Integrated all electronics with an **Arduino Uno** microcontroller.
- Added **HC-05 Bluetooth module** for manual wireless control.
- Wrote Arduino code to control movement directions and conveyor operation.

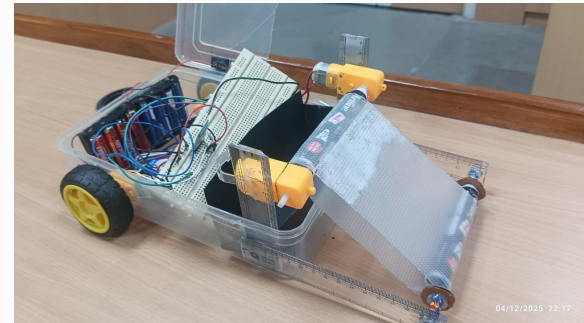
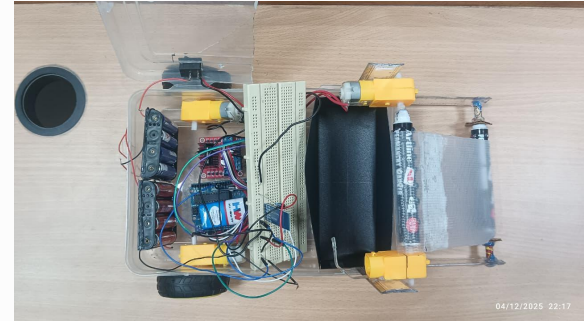
Results and Discussions

Results

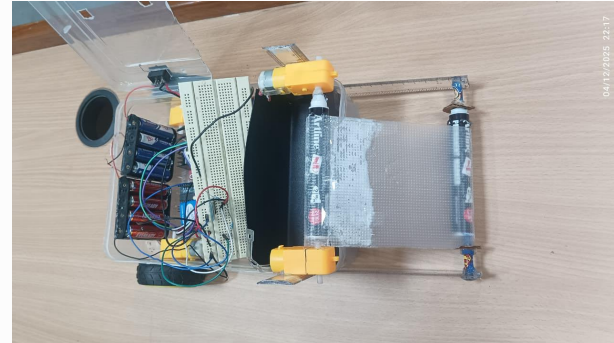
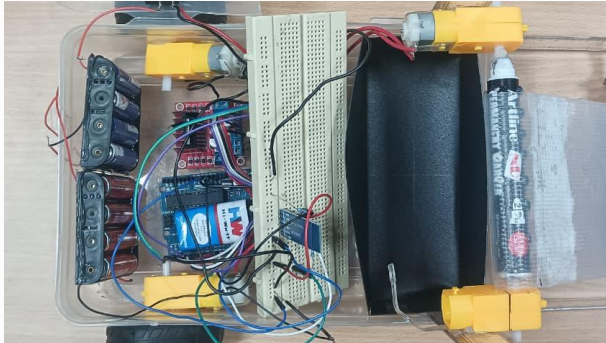
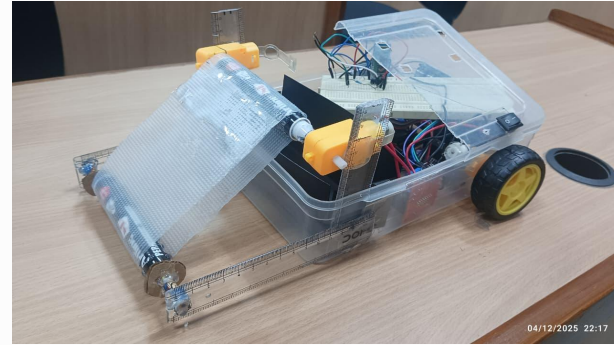
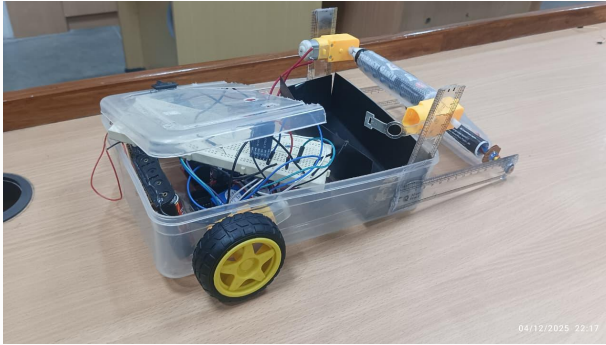
- The robot successfully operated using DC motors controlled through the L298N driver, with both the propeller system and the conveyor mechanism implemented effectively.
- The conveyor mechanism effectively lifted small floating waste such as leaves and paper into the bin.
- Bluetooth control via the HC-05 module worked smoothly, allowing manual operation from a mobile device.

Discussion

- The DC motors and L298N driver provided stable control, enabling smooth movement and conveyor operation.
- The conveyor effectively collected light debris but may require upgrades for heavier waste.
- Bluetooth control worked reliably, though manual operation limits autonomy.
- The prototype shows good potential but needs improvements in automation and load capacity.



Results and Discussions



Conclusion

- The robot successfully moves on water and collects floating waste using a conveyor mechanism.
- Manual Bluetooth control (HC-05) makes it easy to operate and reduces human effort in cleaning.
- Even without automatic sensors, the system works effectively as a low-cost water-cleaning prototype.
- The project provides a good base for future improvements like sensor automation and solar power.

References

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Thanks!